Michelson Interferometry Lab Report

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I. Abstract

Making precise measurements of microscopic quantities is imperative to the advancement of modern physics. Michelson interferometry is a popular interferometer arrangement used to make such measurements using the wave-like nature of light. We use a Michelson interferometer to measure three quantities: (1) the wavelength of a HeNe laser beam and Thallium lamp, (2) the distance between wavelengths of the yellow Mercury doublet, and (3) the index of refraction of a piece of glass. We find a HeNe wavelength of $\lambda = 627 \pm 1$ nm, which is within 5σ of the expected value, and the Thallium lamp wavelength is 530 ± 30nm which agrees with the expected value within 1σ. The distance between the Hg doublet lines is measured as $\Delta\lambda = 2.37 \pm 0.2$ nm which is within 1σ of the expected value. Finally, we measure the refractive index of glass to be $n = 1.37 \pm 0.3$, which agrees within 3σ of the expected $n = 1.51$. This experiment demonstrates the utility of interferometry in making precise measurements of physical quantities.

II. Introduction

Interferometers are powerful instruments which utilize the wave-like behavior of light by analyzing the interference pattern of a split beam of light. Michelson interferometry is a popular arrangement of such tools, becoming famous when it first proved that light did not propagate through a "luminiferous ether" as previously believed by the scientific community (1). The apparatus uses one movable and one stationary mirror to create an interference pattern of a split light source where the two beams travel paths of different lengths (2). Measuring the distance a mirror moves and analyzing the resulting pattern provides for precise measurements of the wavelength of a light source, as first studied, or other measurements such as those we examine here.

Figure 1 shows a basic schematic of the Michelson interferometer. Mirror 1 (M1) and Mirror 2 (M2) are normal to each other and the central beam splitter (BS) is at a 45° angle to both. The BS refracts the light source to two different paths where the beams travel different lengths. M1 is stationary while M2 is free to move, allowing one to create different interference fringe patterns. The final pattern can be viewed on the screen at the top of the diagram. Note that the first beam (1) passes through the BS three times while beam (2) passes through only once.

We use this apparatus to make three different measurements. We first measure the wavelengths of a Helium-Neon laser beam and Thallium lamp by counting fringes by eye. We then measure the separation in wavelength between the yellow Mercury doublet, counting the passing fringes with an automatic photodetector. Finally, we insert a thin piece of glass into the apparatus to measure its refractive index.

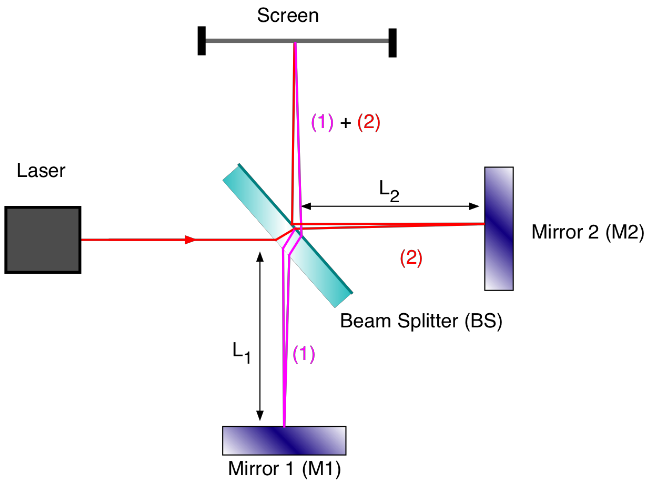

Fig. 1. Schematic of a Michelson interferometer. The laser beam passes through the semi-transparent beam splitter (BS) such that one beam travels to Mirror 1 through the BS three times, while the other travels through only once to Mirror 2 and is then reflected to the screen. The two colors show the paths of the two beams. Image from wanda.fiu.edu/boeglinw.

III. Measuring Light Source Wavelengths

A. Theory

The theory behind this measurement is straightforward and follows directly from the geometry of the experimental apparatus shown in Figure 1. The beams can create three possible interference patterns: hyperbolic fringes (when the mirrors are not parallel and path lengths L1 and L2 are not equal), vertical line fringes (when the mirrors are tilted from normal to the plane of the experiment), and circular fringes (when the apparatus is arranged as shown in Figure 1) (3). In all measurements for this experiment, we use the set up that produces circular fringes.

Suppose that beam one travels a total distance d_1 and beam two travels d_2. Then the beams will constructively interfere and create a bright ring where $|d_1 - d_2| = n\lambda$ where n is an integer, and the destructive interference (dark rings) will occur when $|d_1 - d_2| = n'\frac{\lambda}{2}$, where n' is an odd integer (2). From this, we know that the same pattern will repeat itself when the mirror M2 moves a distance $\lambda/2$. Measuring this distance from the micrometer attached to mirror M2 allows us to solve for the laser beam's wavelength λ.

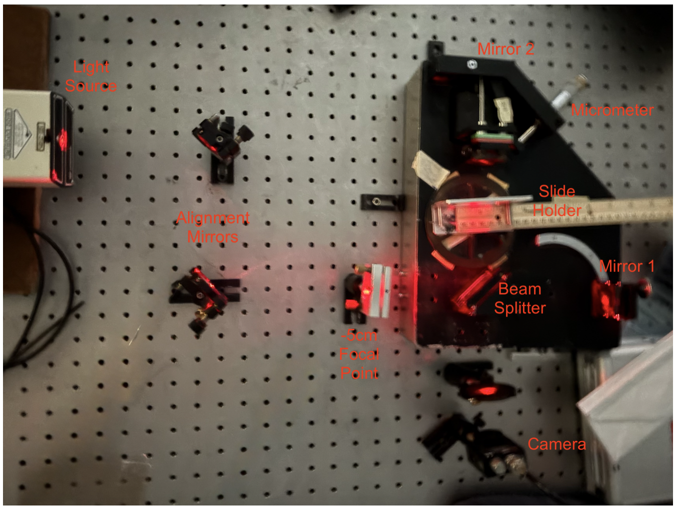

Fig. 2. Experimental set up used for measurements. Note that the set up does not match Figure 1 exactly, but the principles remain the same.

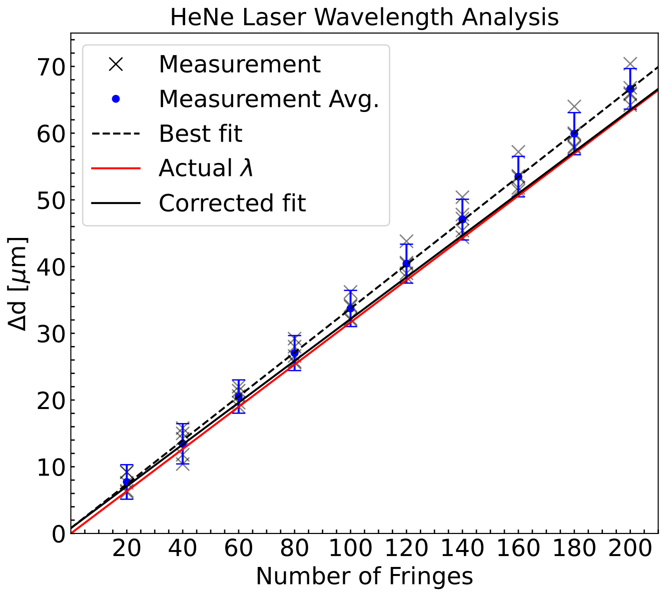

Fig. 3. The change in distance of mirror M2 for every 20 fringes counted by eye. The gray x's represent each of our measurements. The blue circle is the average at each fringe count with an error bar showing the 1σ standard deviation. The dashed line shows the best-fit to our true data, which has a reduced χ^2-fit of 0.10. The solid black line shows the fit with the correction described in this section with a reduced $\chi^2 = 0.99$, and the red line shows the trend given by the expected wavelength.

B. Methods

We arrange the laser interferometer to match the basic configuration in Figure 1. The actual set up is shown in Figure 2. We use two alignment mirrors to level the beam to the plane of the experiment and align it to the center of the beam splitter. We use a -5 cm focal point to diffuse the light so the fringes are easier to see visually. The light travels the paths described in Section III-A and before reaching the camera. The camera is connected with a USB cable to a computer with image analysis software.

We first use an alignment laser to test our set up. We then replace it with the HeNe source. We annotate the camera feed with a vertical line to mark a fixed location, then move the micrometer until we manually count that 20 fringes have passed and record the distance Δd. We repeat this measurement until 200 fringes have passed, and we complete this process a total of five times for 50 total measurements.

We then replace the HeNe laser with a Thallium lamp connected to a high-voltage power supply. The light from the lamp is significantly more diffuse than the laser beam, so we place it closer to the interferometer at the location of the focal point in Figure 2. Instead of counting a constant number of fringes as for the HeNe procedure, we count the number of fringes that pass a fixed point when the micrometer moves a constant distance $\Delta d = 20\mu$m. Because the interference pattern has less contrast than the HeNe case, we record videos on the camera for each 20μm interval and play them back at a 10-25% speed to improve the accuracy of our fringe count. We repeat this process for a total of 15 independent measurements.

C. Analysis and Interpretation

We first calculate the wavelength of the HeNe laser by:

$$\lambda = \frac{2\Delta d}{N} \tag{1}$$

.

Figure 3 shows the change in distance of the path length for the beam traveling to mirror M2 as a function of the total number of fringes that pass through a fixed point. The gray x's show our measurements, and the blue circle and error bar are the average and standard deviation of these measurements for each fringe count. We calculate a weighted best-fit line to these averages, which is shown as the black dashed line. From Equation 1, we calculate a wavelength of 658 ± 1 nm. The error is calculated from the diagonalized covariance matrix for our best-fit parameters. This result is 26σ from the expected value, which is shown as the solid red line, indicating a strong systematic bias in our procedure. Because we counted the fringes by eye, it is extremely likely that we missed fringes that had passed by, resulting in an under-count of N. We assume that we miss 1 in every 20 fringes and apply a cumulative correction to the fringe count. We then re-fit the data and re-calculate the wavelength. With this correction, we find $\lambda = 627 \pm 1$ nm. This result is 6σ lower than the expected value, meaning that we overestimated the total number of missing fringes, but it accounts for a majority of the disagreement. Additional uncertainty in our measurement comes from manually reading the distance from the micrometer, which can only be read within $\pm 1\mu$m.

We complete a similar analysis to calculate the Thallium spectral line wavelength. Table I details our raw data for the number of fringes that pass when M2 is moved through a constant distance of 20μm. We note that even at a 10% playback speed, one of the videos was not smooth enough to provide a reliable fringe count, so we do not include this value in our analysis. For the 14 remaining measurements, we complete the same process as for the HeNe wavelength. Our weighted fit gives a wavelength estimate $\lambda = 580 \pm 30$ nm, which is more than 1σ from the expected value. We

Δd Range ($\pm 1\mu$m)	N_1	N_2	N_3	N_4	N_5
$0-20\mu$m	63	63	69	70	70
$20-40\mu$m	68	72	69	60*	69
$40-60\mu$m	72	60	65	73	71

TABLE I
RAW DATA FOR THE THALLIUM LAMP WAVELENGTH CALCULATION. THE LEFT COLUMN SHOWS THE RANGE OVER WHICH THE MIRROR M2 MOVED AND THE FRINGE COUNTS FOR FIVE INDEPENDENT MEASUREMENTS ARE LISTED. THE STARRED POINT (*) HAD A VIDEO THAT WAS TOO JERKY TO GIVE A RELIABLE COUNT, SO IT WAS NOT INCLUDED IN THE CALCULATION.

N	d_1 ($\pm 1\mu$m)	d_2 ($\pm 1\mu$m)	d_3 ($\pm 1\mu$m)
0	0	0	0
1	84	80	80
2	162	162	160
3	244	242	240
4	324	322	320

TABLE II
CUMULATIVE DISTANCES RECORDED BETWEEN HIGH AND LOW CONTRAST CYCLES USING THE HG LIGHT SOURCE.

find evidence of the same systematic bias in undercounting fringes, and add a 10% cumulative correction for the number of missing fringes. We repeat the fitting procedure and best-fit parameter calculation with the corrected fringe count and find the wavelength is $\lambda = 530 \pm 30$ nm. This result agrees extremely well with the expected value $\lambda = 535$ nm. Our measurement has the same uncertainty in micrometer reading. Interferometers are extremely sensitive to their environment as well, so even slight disturbances like footsteps of breaths near the instrument can disrupt the experiment adding additional sources of systematic error that are difficult to account for. Using an automatically moving interferometer in a closed, controlled environment would reduce these potential errors.

IV. MEASURING SEPARATION OF HG DOUBLET LINES

A. Theory

The Mercury spectrum has two lines in yellow light that are closely spaced, requiring special techniques to measure the difference in their wavelengths. As described in the lab materials (3), two spectral lines with similar wavelengths will have overlapping interference patterns with this experimental set up, resulting in either high or low contrast rings (i.e. clearly distinguishable alternating bright/dark rings, or rings which appear more "washed out"). Thus for a given angle θ, the maximum and minimum will appear on the screen at length d away at $2d\cos(\theta) = (m + 1/2)\lambda$ and $2d\cos(\theta) = m\lambda$, respectively.

If the spectral lines peak at wavelengths λ_1 and λ_2 then we can find their difference $\Delta\lambda$ by $\Delta\lambda = \lambda_2 - \lambda_1$ with $\Delta\lambda/\lambda_1 \ll 1$. Suppose there is a dark ring at angle θ on the screen. Then we know that $2d\cos(\theta) = m_1\lambda_1 = m_2\lambda_2$ and that the next dark fringe appears at $2d'\cos(\theta) = m_1'\lambda_1 = m_2'\lambda_2$, where $m_2' = m_1 + k$ and $m_1' = m_1 + k + 1$ and k is an integer. This yields $2(d' - d)\cos(\theta) = (k + 1)\lambda_1 = k\lambda_2$. Substituting the expression for $\Delta\lambda$ yields:

$$\Delta\lambda = \frac{\lambda_1\lambda_2}{2\Delta d\cos(\theta)} \tag{2}$$

Because the difference between λ_1 and λ_2 is negligible compared the their values, the numerator can be approximated with a term λ_{avg} representing the average of these two wavelengths. The equation is further simplified by picking a fixed point near the center of the interference pattern where $\theta \approx 0$. Thus the final expression we use is:

$$\Delta\lambda = \frac{\lambda_{\text{avg}}^2}{2\Delta d} \tag{3}$$

B. Methods

Our set up is very similar to that described in Section III-B for the Thallium source measurement. Instead of the Thallium source, we use a Mercury lamp with a green glass filter placed in front of the window. This is located directly in front of the beam splitter because the light is diffuse.

Using the camera attached to the computer, we count four full cycles of high/low contrast fringe patterns and record the displacement of the interferometer after each completed cycle. We repeat this process three times for a total of 15 measurements. Our measurements are detailed in Table II.

C. Analysis and Interpretation

The Hg doublet lines occur at $\lambda = 576.9599$ nm and $\lambda = 579.0664$ nm, from which we calculate λ_{avg}. To calculate the typical distance Δd for one cycle, we take each row of Table II and subtract the row before it. This yields 12 measurements of the mirror displacement for a single cycle. We calculate the spectral line separation from Equation 3, then take the mean of these estimates as our measurement.

Our measurement error comes from the manually controlled interferometer. At the beginning and end of each observation, the measurement precision is limited to within $\pm 1/2$ subdivisions of the micrometer, equivalent to 1μm. We propagate this error according to the standard formula:

$$\sigma_{\Delta\lambda}^2 = \left(\frac{\partial\Delta\lambda}{\partial d}\right)^2 \sigma_d^2 \tag{4}$$

where σ_d is the standard deviation of our measurements. Thus we calculate a final measurement for the separation between the Mercury doublet spectral lines: $\Delta\lambda = 2.07 \pm 0.03$ nm. Since we know the actual spectral line wavelengths, the expected value is $\Delta\lambda = 2.11$ nm, which is within 2σ of our estimate.

V. MEASURING REFRACTION INDEX OF GLASS

A. Theory

Following the derivation in Fendley (1982) (4), suppose there is a plate with index of refraction n that we want to measure using a light source with wavelength λ. The beam that passes through the plate of thickness d at an angle ϕ_i to the mirror will be displaced by

$$\frac{2\pi n\overline{PQ}}{\lambda} = \frac{2\pi nd}{\lambda\,\cos(\phi_r)} \tag{5}$$

when going from P to Q and the other beam will be displaced by

$$\frac{2\pi\overline{PS}}{\lambda} = \frac{2\pi d\ \cos(\phi_i - \phi_r)}{\lambda\ \cos^2(\phi_r)} \tag{6}$$

such that the phase difference between the two beams becomes

$$\Delta = 2\frac{2\pi nd}{\lambda\ \cos(\phi_r)} - \frac{2\pi d\ \cos(\phi_i - \phi_r)}{\lambda\ \cos(\phi_r)} \tag{7}$$

where ϕ_r is the angle of refraction and $\sin(\phi_i) = n\cos(\phi_r)$.

If $\phi_i = 0$, then the displacement is $\Delta_o = 2(2\pi nd/\lambda - 2\pi d/\lambda)$. By changing the angle of incidence from $\phi_i = 0$ to ϕ_i, the resulting inference pattern of m fringes can be analyzed such that

$$m = \frac{2d}{\lambda}\left(n\left(\frac{1}{\cos(\phi_r)} - 1\right) + 1 - \frac{\cos(\phi_i - \phi_r)}{\cos(\phi_r)}\right) \tag{8}$$

where n can be solved for numerically by minimizing this equation.

An equivalent relation for measuring the index of refraction is

$$n = \frac{(2t - N\lambda)(1 - \cos\theta)}{2t(1 - \cos\theta) - N\lambda} \tag{9}$$

where N is the number of fringes, θ is the angle the slide was turned through, and d is the thickness of the material (5).

B. Methods

We use the same experimental set up as Figure 2 with the HeNe laser. Instead of a camera which requires counting the fringes by eye, we use a high speed Silicon photodetector to automatically count the fringes. We measure the thickness of a glass slide with a sliding caliper 5 times and take the mean and standard deviation for our thickness and associate uncertainty. The thickness is $t = 1.02 \pm 0.03$ mm. We insert a piece of glass into the slide holder labeled in Figure 2 which is attached to a rotating protractor.

We start the plate at an angle incident to M2 and record the starting position. The slide is parallel to M2 when the protractor reads 4.5 markings, so we subtract this value to make it our 0 point. The protractor is divided into 50 intervals with 10 sub-markings per interval, meaning that our measurement precision is limited to $2\pi/1000$, since we can measure within $\pm 1/2$ subdivision on each side. We then turn the plate through 2 full markings towards the 0 position, recording cumulatively how many fringes pass every $1/2$ marking. We repeat this process three times for a total of 12 measurements, which are detailed in Table III. Because we use an automatic photodetector, we do not record uncertainties with each count but instead propagate their discrepancies to the final value of n.

C. Analysis and Interpretation

We use the angle turned through and number of fringes that pass through a fixed point during that motion to calculate n from Equation 9. We take the mean of each row to and use Equation 9 to calculate a single n value over that interval θ.

We again propagate our error according to the standard prescription, where now several variables have associated measurement uncertainties. For each estimate, we calculate its uncertainty by:

Starting $\theta \pm 0.006$ (rad)	N1	N2	N3
0	0	0	0
0.691	58	56	60
0.628	119	110	109
0.566	115	159	157
0.502	198	208	192

TABLE III
RAW DATA FOR MEASURING THE INDEX OF REFRACTION OF A PLATE OF GLASS. WE RECORD THE STARTING ANGLE OF EACH INTERVAL AND THE TOTAL NUMBER OF FRINGES COUNTED BY THE PHOTODETECTOR IN THAT INTERVAL.

$$\sigma_n^2 = \left(\frac{\partial n}{\partial N}\right)^2 \sigma_N^2 + \left(\frac{\partial n}{\partial \theta}\right)^2 \sigma_\theta^2 + \left(\frac{\partial n}{\partial t}\right)^2 \sigma_t^2 \tag{10}$$

The largest uncertainties come from manually measuring the angle θ, since the number of fringes are recorded with an automatic photodetector and the uncertainty in the thickness is small relative to the measured quantity. For our final measurement, we average over the four estimates and add their associated error in quadrature. We find a refractive index of $n = 1.38 \pm 0.05$. We agree within 3σ of the expected value $n = 1.51$ (6).

VI. CONCLUSION

We have demonstrated the utility of Michelson interferometry to measure physical quantities in three independent mini-experiments. All of our measurements agree within 3σ of the expected value, except for the wavelength of the HeNe laser which suggests a more sophisticated correction is required to correct our systematic error. In general, we were able to measure the quantities of interest remarkably well by utilizing the geometry of the interferometer system. Our measured quantities are restated below:

- The wavelength of the Helium-Neon laser is $\lambda = 627 \pm 1$ nm.
- The wavelength of the Thallium lamp is $\lambda = 530 \pm 30$ nm.
- The separation between the spectral lines of the Mercury doublet is $\Delta\lambda = 2.07 \pm 0.03$ nm.
- The index of refraction of the glass slide is $n = 1.38 \pm 0.05$.

With increasingly complicated physical systems, making precise measurements remains vital to verifying theories and uncovering new properties of materials. Interferometry will continue to be a powerful method for conducting such experiments in physics.

VII. ACKNOWLEDGEMENTS

I would like to thank my lab partners JS and CF for their insights and contributions and great taste in music.



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